

		MATH +maj1@k		GSM8k +maj1@k	
PaLM	8B	1.5	-	4.1	-
	62B	4.4	-	33.0	-
	540B	8.8	-	56.5	-
Minerva	8B	14.1	25.4	16.2	28.4
	62B	27.6	43.4	52.4	68.5
	540B	33.6	50.3	68.5	78.5
LLaMA	7B	2.9	6.9	11.0	18.1
	13B	3.9	8.8	17.8	29.3
	33B	7.1	15.2	35.6	53.1
	65B	10.6	20.5	50.9	69.7

Table 7: **Model performance on quantitative reasoning datasets.** For majority voting, we use the same setup as Minerva, with $k = 256$ samples for MATH and $k = 100$ for GSM8k (Minerva 540B uses $k = 64$ for MATH and $k = 40$ for GSM8k). LLaMA-65B outperforms Minerva 62B on GSM8k, although it has not been fine-tuned on mathematical data.

docstring. The model needs to generate a Python program that fits the description and satisfies the test cases. In Table 8, we compare the pass@1 scores of our models with existing language models that have not been finetuned on code, namely PaLM and LaMDA (Thoppilan et al., 2022). PaLM and LLaMA were trained on datasets that contain a similar number of code tokens.

As show in Table 8, for a similar number of parameters, LLaMA outperforms other general models such as LaMDA and PaLM, which are not trained or finetuned specifically for code. LLaMA with 13B parameters and more outperforms LaMDA 137B on both HumanEval and MBPP. LLaMA 65B also outperforms PaLM 62B, even when it is trained longer. The pass@1 results reported in this table were obtained by sampling with temperature 0.1. The pass@100 and pass@80 metrics were obtained with temperature 0.8. We use the same method as Chen et al. (2021) to obtain unbiased estimates of the pass@k.

It is possible to greatly improve the performance on code by finetuning models on code-specific tokens. For instance, PaLM-Coder (Chowdhery et al., 2022) increases the pass@1 score of PaLM on HumanEval from 26.2% for PaLM to 36%. Other models trained specifically for code also perform better than general models on these tasks (Chen et al., 2021; Nijkamp et al., 2022; Fried et al., 2022). Finetuning on code tokens is, however, beyond the

scope of this paper.

		HumanEval		MBPP	
pass@	Params	@1	@100	@1	@80
LaMDA	137B	14.0	47.3	14.8	62.4
PaLM	8B	3.6*	18.7*	5.0*	35.7*
PaLM	62B	15.9	46.3*	21.4	63.2*
PaLM-cont	62B	23.7	-	31.2	-
PaLM	540B	26.2	76.2	36.8	75.0
LLaMA	7B	10.5	36.5	17.7	56.2
	13B	15.8	52.5	22.0	64.0
	33B	21.7	70.7	30.2	73.4
	65B	23.7	79.3	37.7	76.8

Table 8: **Model performance for code generation.** We report the pass@ score on HumanEval and MBPP. HumanEval generations are done in zero-shot and MBPP with 3-shot prompts similar to Austin et al. (2021). The values marked with * are read from figures in Chowdhery et al. (2022).

3.6 Massive Multitask Language Understanding

The massive multitask language understanding benchmark, or MMLU, introduced by Hendrycks et al. (2020) consists of multiple choice questions covering various domains of knowledge, including humanities, STEM and social sciences. We evaluate our models in the 5-shot setting, using the examples provided by the benchmark, and report results in Table 9. On this benchmark, we observe that the LLaMA-65B is behind both Chinchilla-70B and PaLM-540B by a few percent in average, and across most domains. A potential explanation is that we have used a limited amount of books and academic papers in our pre-training data, i.e., ArXiv, Gutenberg and Books3, that sums up to only 177GB, while these models were trained on up to 2TB of books. This large quantity of books used by Gopher, Chinchilla and PaLM may also explain why Gopher outperforms GPT-3 on this benchmark, while it is comparable on other benchmarks.

3.7 Evolution of performance during training

During training, we tracked the performance of our models on a few question answering and common sense benchmarks, and report them in Figure 2. On most benchmarks, the performance improves steadily, and correlates with the training perplexity of the model (see Figure 1). The exceptions are SIQA and WinoGrande. Most notably, on SIQA,